

24.1 Ultrasound & X-rays

Question Paper

Course	CIEA Level Physics
Section	24. Medical Physics
Topic	24.1 Ultrasound & X-rays
Difficulty	Medium

Time allowed: 70

Score: /52

Percentage: /100

Question 1a

(i)

State what is meant by contrast in an X-ray image.

[1]

(ii)

A parallel beam of X-ray radiation is incident, separately, on samples of bone and of muscle.

Data for the thickness x of the samples of bone and of muscle, together with the linear attenuation coefficients μ of the radiation in bone and in muscle, are given in Table 10.1.

Table 10.1

	x / cm	μ / cm^{-1}
bone	1.5	2.9
muscle	4.0	0.95

Calculate the ratio

$$\frac{\text{intensity of radiation transmitted through the bone}}{\text{intensity of radiation transmitted through the muscle}}.$$

ratio = [2]

[3 marks]

Question 1b

(i)

Explain how ultrasound pulses are used to obtain diagnostic information about internal body structures in medical diagnosis.

[3]

(ii)

Define specific acoustic impedance.

[2]

(iii)

Two media have specific acoustic impedances Z_1 and Z_2 .

State the approximate value of the intensity reflection coefficient at the boundary between the two media when:

- Z_1 is almost equal to Z_2

intensity reflection coefficient =

- Z_1 is very different from Z_2 .

intensity reflection coefficient =

[2]

[7 marks]

Question 2a

(i)

Explain how X-rays are produced for use in medical diagnosis

[3]

(ii)

Discuss **one** advantage and **one** disadvantage of using CT scanning over X-ray imaging.

[2]

[5 marks]**Question 2b**

An X-ray image is taken of the skull of a patient. Another patient has a CT scan taken of their entire head.

Describe how the image produced during CT scanning differs from the image produced by X-ray imaging.

[4 marks]

Question 2c

Ultrasound is thought to be more suitable than X-rays for scanning an unborn foetus.

(i)

Describe how ultrasound imaging can be used to produce an image of an unborn foetus.

[4]

(ii)

When obtaining the ultrasound image of an unborn foetus, a coupling gel is used.

Explain why a coupling gel is needed and state the property of the gel that ensures a good quality image.

[2]

[6 marks]

Question 2d

Some values of linear attenuation coefficient of different media are shown in Table 1.1

Table 1.1

medium	attenuation coefficient / cm^{-1}
soft tissue	0.23
blood	0.03
fat	0.48

- (i)
State what is meant by the attenuation of an ultrasound wave.
- (ii)
A parallel beam of ultrasound passes through a medium.
The incident intensity I_0 is reduced to $0.75 I_0$ on passing through a thickness of 0.094 m of the medium.
Use Table 1.1 to determine the medium that the ultrasound passed through.

[1]

[4]

[5 marks]

Question 3a

Fig. 1.1 shows a diagram of a piezoelectric transducer.

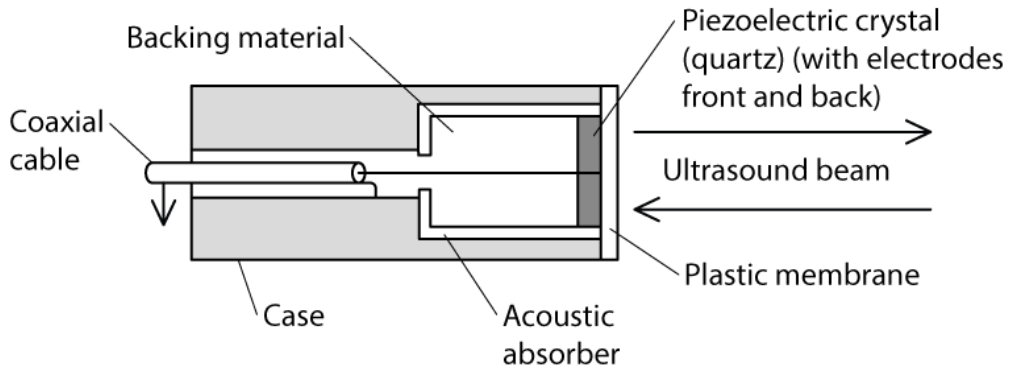


Fig. 1.1

With reference to Fig. 1.1:

- (i)
Describe the process by which the transducer produces a pulse of ultrasound. [4]
- (ii)
Explain how short pulses of ultrasound are produced and state why they are essential. [2]
- (iii)
Explain how the received signals are detected. [2]

[8 marks]

Question 3b

Two patients arrive at a hospital. One patient is a child with a broken arm, the other is a pregnant woman coming in for a check-up for her unborn foetus.

Over the course of the diagnosis and treatment of the child's broken arm, several images of the arm are required.

Similarly, to check the progress of a woman's pregnancy, several images of the foetus are required.

For each scenario, suggest the most appropriate imaging technique to use and give **two** reasons for the choice.

[4 marks]

Question 3c

Ultrasound is incident on a boundary between two materials. Some of the ultrasound is reflected at the boundary and the rest is transmitted across the boundary.

The acoustic impedance of air is $4.29 \times 10^2 \text{ kg m}^{-2} \text{ s}^{-1}$

The acoustic impedance of coupling gel is $1.48 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$

The acoustic impedance of skin is $1.65 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$

Calculate the percentage of the ultrasound which would be transmitted into the skin when incident on

(i)
an air–skin boundary,

[2]

(ii)
a gel–skin boundary.

[2]

[4 marks]

Question 3d

An X-ray image showing the cross-section of the bone and muscle is shown in Fig. 1.1.

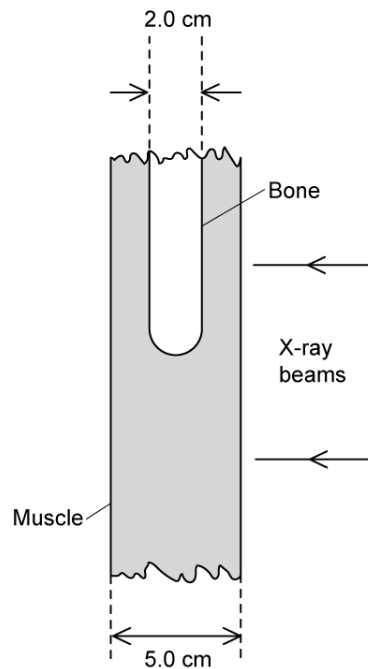


Fig. 1.1

Parallel X-ray beams of intensity I_0 are incident on the muscle and bone as shown. The emergent beam after passing through muscle alone has an intensity of I_m and an intensity of I_{bm} after passing through the bone and the muscle.

(i)

Determine, in terms of I_0 , the intensities I_m and I_{bm} of the emergent X-rays.

The linear attenuation coefficient of bone is 2.90 cm^{-1}

The linear attenuation coefficient of muscle of 0.95 cm^{-1} .

[4]

(ii)

Suggest why, on an X-ray plate, the contrast between bone and muscle is much greater than the contrast between fat and muscle.

The linear attenuation coefficient of fat is 0.90 cm^{-1} .

[2]

[6 marks]

Model Answers: Medium

1a

(a)

(i) State what is meant by contrast in an X-ray image:

- Contrast means a difference in the amount of blackening in the image; [1 mark]

(ii) Calculate the ratio $\frac{\text{intensity of radiation transmitted through the bone}}{\text{intensity of radiation transmitted through the muscle}}$:

- $I = I_0 e^{-\mu x}$ [1 mark]
- $I_b = I_0 e^{-2.9 \times 1.5}$
- $I_m = I_0 e^{-0.95 \times 4}$
- $\frac{I_b}{I_m} = \frac{I_0 e^{-2.9 \times 1.5}}{I_0 e^{-0.95 \times 4}} = \frac{e^{-2.9 \times 1.5}}{e^{-0.95 \times 4}} = 0.577 = 0.58 \text{ (2 s.f.)}$ [1 mark]

[Total: 3 marks]

1b

(b)

(i) How ultrasound pulses are used to obtain diagnostic information:

- Ultrasound pulses are reflected (from boundaries between different densities of medium); [1 mark]
- The time between the transmission and detection (of the reflected pulse) indicates the depth (of the boundary); [1 mark]
- The intensity (of the reflected pulse) indicates the nature of the boundary; [1 mark]

(ii) Specific acoustic impedance is defined as:

- The product of the density (of the medium) and the speed; [1 mark]
- Of sound in the medium; [1 mark]

(iii) State the approximate value of the intensity reflection coefficient at the boundary between the two media when:

Z_1 is almost equal to Z_2 :

- $\alpha \approx 0$; [1 mark]

Z_1 is very different from Z_2 :

- $\alpha \approx 1$; [1 mark]

[Total: 7 marks]

2a

(a)

(i) X-rays are produced by:

- Accelerating electrons (through a high p.d.); [1 mark]
- Electrons are incident on a metal target; [1 mark]
- (When electrons hit the target) X-rays are produced as the electrons decelerate; [1 mark]

(ii) One advantage and disadvantage of using CT scanning is:

One advantage is:

Any **one** from:

- (CT scans produces) more detailed (3D) image which is better for diagnosis; [1 mark]
- (CT scans) can distinguish between different structures with similar attenuation coefficients as X-ray scans just detect a 'shadow'; [1 mark]

One disadvantage is:

Any **one** from:

- Patients are exposed to a higher dose of ionising radiation which could cause cancer / damage cells; [1 mark]
- Usually requires contrast media which can have negative / adverse side effects; [1 mark]

[Total: 5 marks]

2b

(b) The image produced during CT scanning differs from the image produced by X-ray imaging by:

Max **four** from:

- An X-ray image is flat / 2D; [1 mark]
- CT scan takes many images of a section / slice at different angles; [1 mark]
- The images are combined to build up an image of a section / slice through the body; [1 mark]
- The process is repeated for different sections / slices; [1 mark]
- These images are combined together to build a 3D image; [1 mark]

[Total: 4 marks]

2c

(c)

(i) Ultrasound can produce an image of an unborn foetus by:

Any **four** from:

- Moving a transducer across the surface of the skin; [1 mark]
- While the transducer emits (short) pulses of ultrasound waves; [1 mark]
- (Ultrasound) waves are reflected at boundaries where acoustic impedance changes; [1 mark]
- The time (delay) for the pulse to return is measured; [1 mark]

- Time delay gives information about the depth / position of the boundary; [1 mark]
- The intensity of the reflection gives information about the nature of the boundary; [1 mark]

(ii) A coupling gel is needed because:

Any **two** from:

- The acoustic impedance of air is very different to skin **OR** gel is used between the probe and the skin to exclude air; [1 mark]
- Gel has an acoustic impedance equal / similar to the skin / soft tissue; [1 mark]
- This enables maximum transmission / greatly increases transmission into the body **OR** this minimises reflection / greatly reduces reflection at the body boundary; [1 mark]

[Total: 6 marks]

2d

(d)

(i) Attenuation of an ultrasound wave is:

- The loss of intensity / amplitude / power (of the wave); [1 mark]

(ii) Determine the medium that the ultrasound passed through:

List the known quantities:

- Intensity, $I = 0.75 I_0$
- Medium thickness, $x = 0.094 \text{ m} = 9.4 \text{ cm}$

Determine the linear attenuation coefficient:

- Attenuation of ultrasound: $I = I_0 e^{-\mu x}$ [1 mark]
- $0.75 I_0 = I_0 e^{-9.4\mu} \Rightarrow 0.75 = e^{-9.4\mu}$ [1 mark]

- Attenuation of ultrasound: $I = I_0 e^{-\mu x}$ [1 mark]
- $0.75I_0 = I_0 e^{-9.4\mu} \Rightarrow 0.75 = e^{-9.4\mu}$ [1 mark]
- $\ln(0.75) = -9.4\mu \Rightarrow \mu = -\frac{\ln(0.75)}{9.4}$
- $\mu = 0.0306 = 0.03 \text{ cm}^{-1}$ [1 mark]

Conclusion:

- The medium is blood; [1 mark]

[Total: 5 marks]

3a

(a)

(i) The transducer produces a short pulse of ultrasound by:

Any **four** from:

- Connecting the electrodes to an alternating / high frequency emf / voltage / p.d.; [1 mark]
- The frequency of the alternating p.d. is equal to the resonant frequency of the crystal (which is above 20 kHz); [1 mark]
- (Hence) the crystal expands and contracts / vibrates at the same frequency as the alternating p.d.; [1 mark]
- The vibration produces ultrasound / pressure waves in the crystal / plastic membrane; [1 mark]
- The backing material damps the oscillations of the crystal / stops the vibration of crystal; [1 mark]

(ii) Short pulses of ultrasound are produced by:

- Short applications of alternating p.d. (which stop the crystal oscillating); [1 mark]

Short pulses are essential because:

- The transmission must stop so that the reflected pulse can be received **OR** to ensure the transmitted and reflected pulses do not interfere; [1 mark]

(iii) Received signals are detected by:

- The probe acts as a receiver and a transmitter, so the received signal causes the crystal to vibrate; [1 mark]
- The vibration of the crystal induces an alternating p.d. (which is detected); [1 mark]

[Total: 8 marks]

3b

(b)

For a broken arm, x-ray imaging is the most appropriate because:

Any **two** from:

- X-ray imaging provides a good contrast between bone and the surrounding tissue; [1 mark]
- X-ray creates a sharp / high resolution image (compared to ultrasound); [1 mark]
- X-rays pass straight through soft tissue, whereas ultrasound does not; [1 mark]

For a fetal scan, ultrasound imaging is the most appropriate because:

Any **two** from:

- Ultrasound is safer / non-ionising (for the mother and child); [1 mark]
- Ultrasound can detect changes in tissue type; [1 mark]
- Ultrasound transmission / detection allows imaging in real-time; [1 mark]

[Total: 4 marks]

3c

(c)

List the known quantities:

- Acoustic impedance of air, $Z_a = 4.29 \times 10^2 \text{ kg m}^{-2} \text{ s}^{-1}$
- Acoustic impedance of coupling gel, $Z_g = 1.48 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$
- Acoustic impedance of skin, $Z_s = 1.65 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$

(i) Calculate the % transmitted into the skin for the air–skin boundary:

- Intensity reflection equation: $\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
- $\frac{I_R}{I_0} = \frac{(Z_s - Z_a)^2}{(Z_s + Z_a)^2} = \frac{(1.65 \times 10^6 - 4.29 \times 10^2)^2}{(1.65 \times 10^6 + 4.29 \times 10^2)^2}$
- Fraction reflected: $\frac{I_R}{I_0} = 0.99896$ [1 mark]
- % transmitted = $(1 - 0.99896) \times 100\% = 0.1\%$ [1 mark]

(ii) Calculate the % transmitted into the skin for the gel–skin boundary:

- $\frac{I_R}{I_0} = \frac{(Z_s - Z_g)^2}{(Z_s + Z_g)^2} = \frac{(1.65 \times 10^6 - 1.48 \times 10^6)^2}{(1.65 \times 10^6 + 1.48 \times 10^6)^2}$
- Fraction reflected: $\frac{I_R}{I_0} = 2.9499 \times 10^{-3}$ [1 mark]
- % transmitted = $(1 - 2.9499 \times 10^{-3}) \times 100\% = 99.7\%$ [1 mark]

[Total: 4 marks]

Since the equation has the difference in acoustic impedance **squared**, it doesn't actually matter which boundary you use as Z_1 and Z_2 , as $(Z_2 - Z_1)^2 = (Z_1 - Z_2)^2$. Therefore, focus on making sure you've put in the correct numbers into your calculator and **squared** them too, as this is easy to forget!

3d

(d)

(i) Calculate the intensities of the emergent X-rays:

List the known quantities:

- Attenuation coefficient in bone, $\mu_b = 2.90 \text{ cm}^{-1}$
- Thickness of bone, $x_b = 2.0 \text{ cm}$
- Attenuation coefficient in muscle, $\mu_m = 0.95 \text{ cm}^{-1}$
- Thickness of muscle, $x_m = 5.0 \text{ cm}$

Determine the intensity of X-rays transmitted through muscle alone:

- Attenuation of X-rays: $I = I_0 e^{-\mu x}$ [1 mark]
- Intensity transmitted through muscle alone: $I_m = I_0 e^{-\mu_m x_m}$
- $I_m = I_0 e^{-0.95 \times 5.0} = 8.7 \times 10^{-3} I_0$ [1 mark]

Determine the intensity of X-rays transmitted through muscle and bone:

- Intensity transmitted through bone: $I_b = I_0 e^{-\mu_b x_b}$
- Intensity transmitted through muscle and bone: $I_{bm} = I_0 (e^{-\mu_b x_b}) \times (e^{-\mu_m \Delta x})$
- Where $\Delta x = x_m - x_b = 5.0 - 2.0 = 3.0 \text{ cm}$
- $I_{bm} = I_0 (e^{-2.90 \times 2.0}) \times (e^{-0.95 \times 3.0})$ [1 mark]
- $I_{bm} = 1.8 \times 10^{-4} I_0$ [1 mark]

(ii) The contrast between bone and muscle is much greater than between fat and muscle because:

Any **two** from:

- Contrast depends on the difference in attenuation; [1 mark]

- Thickness of bone, $x_b = 2.0 \text{ cm}$
- Attenuation coefficient in muscle, $\mu_m = 0.95 \text{ cm}^{-1}$
- Thickness of muscle, $x_m = 5.0 \text{ cm}$

Determine the intensity of X-rays transmitted through muscle alone:

- Attenuation of X-rays: $I = I_0 e^{-\mu x}$ [1 mark]
- Intensity transmitted through muscle alone: $I_m = I_0 e^{-\mu_m x_m}$
- $I_m = I_0 e^{-0.95 \times 5.0} = 8.7 \times 10^{-3} I_0$ [1 mark]

Determine the intensity of X-rays transmitted through muscle and bone:

- Intensity transmitted through bone: $I_b = I_0 e^{-\mu_b x_b}$
- Intensity transmitted through muscle and bone: $I_{bm} = I_0 (e^{-\mu_b x_b}) \times (e^{-\mu_m \Delta x})$
- Where $\Delta x = x_m - x_b = 5.0 - 2.0 = 3.0 \text{ cm}$
- $I_{bm} = I_0 (e^{-2.90 \times 2.0}) \times (e^{-0.95 \times 3.0})$ [1 mark]
- $I_{bm} = 1.8 \times 10^{-4} I_0$ [1 mark]

(ii) The contrast between bone and muscle is much greater than between fat and muscle because:

Any **two** from:

- Contrast depends on the difference in attenuation; [1 mark]
- The difference between μ of muscle and fat is small (hence poor contrast); [1 mark]
- The difference between μ of bone and soft tissue / fat is large (hence good contrast); [1 mark]

[Total: 6 marks]

Remember in part (i) when the X-rays travel through the muscle and bone, it is